

FORUM:	United Nations Office for Outer Space Affairs
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Introduction

Commercial extraction refers to harvesting natural resources for financial gain. With the rapid development of space technology, resources such as minerals, metals, and ice have attracted increasing interest from Member States and private companies. Indeed, in 2024, the commercial sector accounted for 78% of the global space economy, underscoring the role of private companies in the space sector. However, this growth may create legal disputes, environmental damage, and unequal access to space resources. As current space treaties do not clearly regulate the ownership of extracted resources, different interpretations may lead to conflicts among space actors. Existing measures remain non-binding, while technological developments continue to outpace international regulations. Therefore, establishing a multilateral regulatory framework is crucial to ensuring the sustainable use of space resources.



Figure 1: Harrison Schmitt collecting lunar samples during Apollo 17

Background

The legal foundation of space resource activities began with the Outer Space Treaty, which entered into force in 1967. Developed during the Cold War, the treaty established that outer space should be used for peaceful purposes and for the benefit of all countries. Article II prohibited states from claiming sovereignty over the Moon or other celestial bodies. However, the treaty did not specifically address whether resources could be owned after extraction because commercial space mining was not technologically feasible when the treaty was drafted.

In 1969, Apollo 11 astronauts collected and returned the first lunar rock and soil samples. Although the samples were collected for scientific rather than commercial purposes, the mission demonstrated that extraterrestrial resources could be physically extracted and transported. As space technology advanced, private companies became increasingly involved in activities that had previously been led primarily by governments. This raised questions about whether private actors could extract space resources without violating the Outer Space Treaty.



Figure 2: The Soviet Union, the United Kingdom and the United States signing the Outer Space Treaty in 1967

In response, the United States enacted the Commercial Space Launch Competitiveness Act in 2015, granting its citizens rights over the space resources they obtained. Likewise, Luxembourg introduced similar legislation in 2017, followed by other states. Nevertheless, these domestic laws differed and were not based on a detailed international standard. Later, in 2020, the Artemis Accords introduced principles concerning space resource extraction and stated that extraction did not necessarily constitute

national appropriation. Still, the Accords remained non-binding and applied only to participating states.

Consequently, these developments left several issues unsolved, particularly regarding the ownership of extracted resources and the regulation of private companies. Additional concerns arose over potential environmental damage to celestial bodies and unequal access to limited resources. While previous agreements laid the foundation for regulating the commercial extraction of space resources, remaining legal gaps highlighted the need for further international discussion on the sustainable use of space resources.

Problems Raised

Regulatory Fragmentation and Jurisdictional Loopholes

The absence of a binding international standard allows Member States to regulate commercial space resource extraction through their own domestic laws. Although the Outer Space Treaty prohibits states from claiming sovereignty over the Moon and other celestial bodies, it does not clearly determine whether private companies may own resources after extraction. Its vague language has consequently allowed states to establish different legal interpretations and doctrines. The United States, Luxembourg, the United Arab Emirates, and Japan have introduced domestic legislation recognizing certain rights over extracted resources. However, these laws differ in their licensing procedures, environmental

requirements, and methods of corporate supervision. This raises uncertainty regarding which state has jurisdiction over extraction activities and which legal doctrine should apply when disputes involve several states or private companies.

Differences among domestic laws may incentivize companies to register their operations in jurisdictions with less restrictive regulatory requirements. Although the commercial space-mining industry remains at an early stage, this potential regulatory arbitrage could allow companies to exploit regulatory loopholes and avoid stricter environmental, financial, or reporting obligations. It may also encourage states to weaken their standards to attract foreign investment, potentially creating unequal competition and a regulatory race to the bottom. Without international standardization, fragmented domestic laws may hinder corporate accountability and exacerbate legal and economic disputes among Member States.

Concentration of Capital and Technology



Figure 3: Share and number of operational satellites by country in 2022

Commercial space resource extraction requires considerable infrastructure. Hence, only a limited number of technologically advanced states and private companies currently possess the capacity to participate effectively in the industry. These actors may gain early access to resource-rich sites

and establish significant control over the developing market. Such advantages could allow them to influence resource prices, technical standards, and conditions of access. Furthermore, the control of important patents, extraction equipment, and transportation systems by a small number of corporations may hinder the entry of smaller companies and further concentrate economic power.

Developing countries are particularly vulnerable to this inequality. The Outer Space Treaty establishes the doctrine that outer space should be explored and used for the benefit and in the interests of all countries, regardless of their level of economic or scientific development. Nevertheless, its language remains ambiguous regarding how the benefits of commercial extraction should be distributed. States without sufficient technology may therefore be excluded from both commercial opportunities and international decision-making. This could exacerbate existing technological and economic inequalities while concentrating the benefits of outer space activities among a small group of states and corporations.

Environmental Damage and Harmful Interference

Unregulated extraction may cause lasting damage to the environments of the celestial bodies. Mining operations may disturb surface materials, disperse dust and debris, contaminate scientifically significant areas, or consume limited deposits of water ice before their environmental importance is fully understood. Article IX of the Outer Space Treaty requires states to avoid harmful contamination and interference, but it does not provide detailed environmental standards specifically tailored to commercial extraction. The lack of clarity of this provision makes it difficult to determine the permissible level of environmental damage, the appropriate assessment procedures, and the authority responsible for imposing penalties.

Resource extraction may also interfere with the activities of other institutions. A private company operating near a resource-rich or strategically valuable site could restrict practical access through its equipment, dust production, communication systems, or designated safety zones, even without making a formal claim of sovereignty. Such interference may hinder future scientific research and prevent other actors from accessing the same area. These environmental and scientific repercussions may be especially severe for universities, smaller companies, and developing states with limited opportunities to conduct space missions.

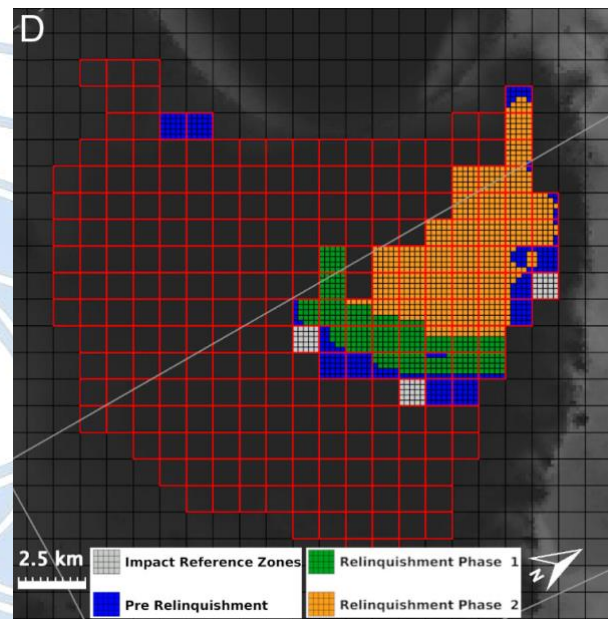


Figure 4: Proposed phased relinquishment zones for lunar mining

International Actions

Outer Space Treaty

The Outer Space Treaty, adopted by the United Nations General Assembly in 1966 and entering into force in 1967, established the fundamental legal framework governing outer space activities. Often regarded as the “constitution of international space law,” it introduced principles that continue to guide subsequent space agreements and national legislation. Article I declares that the exploration and use of outer space should be carried out for the benefit and in the interests of all countries, while Article II prohibits national appropriation through claims of sovereignty, occupation, or other means. The treaty also requires states to supervise the activities of their private entities and avoid harmful contamination and interference.

The treaty has successfully maintained a widely accepted foundation for peaceful space exploration and state responsibility. However, it was drafted before commercial resource extraction became technologically feasible and does not explicitly determine whether resources may be privately owned after extraction. It also provides no detailed licensing, benefit-sharing, or environmental assessment mechanism for commercial mining. Consequently, different interpretations of its broad principles have allowed states to develop conflicting domestic laws. Although the treaty remains the most significant doctrine of international space law, its ambiguous language hinders its direct application to modern commercial extraction.

Moon Agreement

The Moon Agreement was adopted by the United Nations General Assembly in 1979 and entered into force in 1984. It expanded upon the Outer Space Treaty by addressing the Moon and its natural resources more directly. Article 11 identifies the Moon and its resources as the “common heritage of mankind” and calls for an international regime to govern their exploitation once extraction becomes feasible. The proposed regime was intended to ensure the orderly and safe development of lunar



Figure 5: Countries participating in the Moon Treaty as of August 2023

resources, the rational management of those resources, and an equitable distribution of their benefits among Member States. Nevertheless, the Moon Agreement received limited international support, particularly from states possessing advanced space capabilities. Concerns regarding the meaning of “common heritage,” the distribution of economic benefits, and possible restrictions on private investment hindered its widespread acceptance. As a result, the agreement has not developed into a universal framework for commercial resource extraction. Although it represents one of the clearest international endeavours to regulate extraterrestrial resources, its limited participation

prevents it from establishing consistent standards across the international community.

Artemis Accords

The Artemis Accords were established in 2020 by the United States and seven other founding nations to provide practical principles for civil space exploration. Grounded in the Outer Space Treaty, the Accords address peaceful purposes, transparency, scientific data sharing, resource utilization, the

preservation of heritage sites, and the prevention of harmful interference. Regarding space resources, they state that extraction and utilization do not inherently constitute national appropriation under the Outer Space Treaty. They also introduce temporary safety zones intended to improve coordination and prevent interference between operations. As of July 2026, 70 states had signed the Accords.

The Artemis Accords have expanded international discussion and provided operational guidance for states participating in lunar exploration. However, they are non-binding and apply only to their signatories. They were also developed outside the UN treaty-making process, leading to concerns that their principles may reflect the interests of participating states rather than a universal consensus. Furthermore, the Accords do not establish a central regulatory authority or a detailed mechanism for environmental protection and equitable benefit-sharing. Therefore, although they ameliorate some uncertainties surrounding space resource activities, they do not eliminate the need for a binding and internationally standardized framework.

Key Players

United Nations Committee on the Peaceful Uses of Outer Space (UN COPUOS)

The United Nations Committee on the Peaceful Uses of Outer Space is the principal international forum for discussing the peaceful exploration and use of outer space. Established by the UN General Assembly in 1959, COPUOS contributed to the development of the five United Nations treaties on outer space, including the Outer Space Treaty and the Moon Agreement. Its Legal Subcommittee examines international space law, while its Scientific and Technical Subcommittee addresses the safety and long-term sustainability of space activities. The United Nations Office for Outer Space Affairs supports the Committee and its subcommittees by providing secretariat services and promoting international cooperation.

COPUOS has established a Working Group on Legal Aspects of Space Resource Activities to examine possible legal models for the exploration, exploitation, and utilization of space resources. This enables Member States to exchange views on ownership, environmental protection, benefit-sharing, and the responsibilities of private companies. As COPUOS generally operates through consensus, it provides both developed and developing states with opportunities to participate in forming international standards. However, disagreements among Member States and its lack of direct enforcement authority may hinder the rapid adoption of binding regulations.

United States

The United States is one of the most influential actors in commercial space resource extraction. Through the Commercial Space Launch Competitiveness Act of 2015, it recognized the rights of US citizens to possess, use, transport, and sell space resources they obtain, while formally rejecting claims of sovereignty over celestial bodies. The United States also led the establishment of the Artemis



Figure 6: A session of the UN Committee on the Peaceful Uses of Outer Space

Accords in 2020, which state that resource extraction and utilization can be conducted consistently with the Outer Space Treaty. These policies demonstrate the country's support for private ownership, commercial investment, and the practical use of lunar resources. NASA supports this approach by cooperating extensively with private companies. Through the Commercial Lunar Payload Services initiative, NASA contracts American companies such as Intuitive Machines, Firefly Aerospace, and Blue Origin to deliver scientific and technological payloads to the Moon. Some missions have tested technologies related to lunar soil and water ice, which may contribute to future resource utilization. Nevertheless, the United States' domestic recognition of ownership and its leadership of the non-binding Artemis Accords have raised concerns that American standards could influence international practice without universal agreement. Therefore, the United States remains central to negotiations concerning the balance between commercial development and multilateral regulation.

China

China is another major actor due to its advanced lunar exploration programme and growing technological capacity. Led by the China National Space Administration, the Chang'e programme has conducted lunar orbiting, landing, and sample-return missions. Chang'e 5 returned lunar samples in 2020, while Chang'e 6 returned samples from the far side of the Moon in 2024. These missions have strengthened China's capacity to examine lunar materials and develop technologies that could support future resource utilization.

In 2021, China and Russia signed a memorandum of understanding to cooperate on the International Lunar Research Station, a planned scientific base on the lunar surface or in lunar orbit. The project is intended to conduct lunar exploration, scientific experiments, technical verification, and the utilization of lunar resources, and has been presented as open to other international partners. The ILRS

represents a major cooperation framework separate from the United States-led Artemis system. Although competition between these frameworks may accelerate technological development, it could also exacerbate regulatory fragmentation and geopolitical tension. China's participation in COPUOS discussions is therefore essential to ensuring that future international standards are not limited to one political or technological bloc.

Possible Solutions

Establishing an International Regulatory Framework

Considering the differences among domestic space laws, Member States could develop an international framework establishing minimum standards for commercial resource extraction. The framework may clarify ownership, licensing, jurisdiction, corporate responsibility, and dispute resolution while remaining consistent with the Outer Space Treaty. Common standards could reduce regulatory loopholes and prevent companies from relocating to jurisdictions with weaker regulations. However, the framework should allow some national flexibility to encourage technological development and private investment.

Capacity-Building and Data-Sharing for Developing Countries

Member States could strengthen capacity-building programmes to enable developing countries to participate in space resource activities. These endeavors may include technical training, research partnerships, financial assistance, and access to scientific data collected during exploration missions. Transparent data-sharing could reduce technological inequality and allow states without independent space programmes to contribute to research and decision-making. Such measures may also ensure that the benefits of space resources are distributed more equitably.

Strengthening Environmental Standards and Safety Zones

UN COPUOS could cooperate with relevant UN environmental bodies to develop environmental standards for commercial extraction. These standards may include environmental assessments, contamination prevention, restoration responsibilities, and clearer rules for safety zones. Safety zones could be re-established as temporary areas designed to prevent harmful interference rather than enable territorial claims. Member States may also consider making essential environmental obligations legally binding and establishing appropriate repercussions for serious violations. However, Member States should determine how these measures would be monitored and enforced.

Key Terms

Benefit sharing

The idea that the benefits gained from space resources should not belong only to a few countries or companies, but should be shared fairly.

Celestial bodies

Natural objects in space, such as the Moon, planets, and asteroids; these are the places where space resources may be found.

Jurisdiction

the legal authority of a country to regulate space activities carried out by its citizens, companies, or registered space objects.

Sovereignty

The authority of a country over its own territory.

Multilateralism

A way of making decisions through discussion and cooperation among many countries, rather than by one country alone.

Ownership

The question of who has the right to own or use resources taken from space, especially when private companies are involved.

Remote sensing

The use of satellites or other tools to study celestial bodies from a distance before landing or extracting resources.

Safety zone

A protected area around a space activity site, used to prevent accidents or interference

Space resource activities

All activities related to finding, studying, extracting, or using resources in space.

Space resource extraction

The process of taking resources, such as water, metals, or minerals, from celestial bodies.

Space resources

Useful materials found in space, such as water, metals, and minerals.

Spacefaring nations

Countries that have the technology and ability to launch spacecraft or carry out space activities.

Sustainable use

The responsible use of space resources so that they are not wasted or damaged, and can still be used in the future.



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